AAKASH SANGANI

AI governance and auditable AI research

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RESEARCH INTERESTS

Auditable and accountable AI systems. Current threads: inferring the revealed objectives of deployed AI agents from their behaviour; the reliability of the evaluators we use to make safety claims; algorithmic fairness in Indian credit markets; and conformity assessment and incident reporting under the EU AI Act and California's SB-53.

EDUCATION

B.Tech, Information Technology 2019 – 2023

D. J. Sanghvi College of Engineering, University of Mumbai CGPA 8.90 / 10

- Credit-weighted across eight semesters. Merit-based scholarship covering 100% of tuition.
- Relevant coursework: Probability & Statistics (O, 93%), Statistical Analysis (A+, 81%), Artificial Intelligence (A+, 84%), Advanced Data Structures (O, 95%), Design and Analysis of Algorithms (A+, 83%), Discrete Structures (A+, 84%).

RESEARCH AND INDEPENDENT WORK

Inverse-Objective Regulator `github.com/aakash2410/Inverse-Objective-Regulator`

Black-box AI-safety auditing system; sole author of the accompanying arXiv paper.

- Infers a deployed agent's revealed objective from its tool-call traces via Bayesian inverse reinforcement learning, scores divergence from its declared purpose, and emits regulator-legible evidence mapped to the EU AI Act, OWASP, MITRE ATLAS and NIST AI RMF.
- Released an open 8-agent benchmark spanning agentic safety failure modes (reward hacking, power-seeking, sycophancy, privacy leakage, safety-skipping). A deterministic open cross-encoder critic recovers planted objective divergence on 6 of 8 agents (mean r = 0.55), beating a frontier LLM-as-judge (mean r 0.08 → 0.43) with near-zero variance and no API cost.
- Full Python implementation: Bayesian IRL, LLM and decision-model judges, a reproducible gym, and JSON/SARIF reporting.

Credit Equity Audit of India (NSSO 77th Round) `github.com/aakash2410/CreditAudit-India`

Independent fairness audit of household credit access and allocation.

- Two-stage, Heckman-style deep learning framework separating access to the credit market from the allocation of formal versus informal credit among households already in it.
- Female-headed households had about 59% of the probability of entering the credit market compared with male-headed households (disparate impact 0.59), against the conventional 0.8 threshold.
- Applied IBM AIF360 adversarial debiasing and SHAP to test for proxy discrimination, reaching near-ideal disparate impact at ~73.2% accuracy. Documented the limits of the design: self-reported survey data, collider bias risk, and SHAP as a diagnostic of the model rather than causal evidence.

The Specialist, on the Suno Sutra edge platform github.com/aakash2410/SpecialistAISunoSutra

Grounded offline retrieval with explicit abstention, on a government-backed edge device.

- Built and packaged a subject-expert module for Suno Sutra, the open-source offline AI device from Digital India Bhashini (MeitY) and Current AI, which runs vision, ASR, translation and TTS fully on-device across 22 Indic languages.
- Answers are grounded in approved DIKSHA content and carry a citation to chapter, section and page. Two gates enforce abstention: retrieval refuses below a similarity threshold before an LLM call is spent, and the grounded prompt makes the model emit a no-answer signal the pipeline converts into a refusal.
- multilingual-e5-small embeddings with a deterministic, dependency-free TF-IDF fallback fitted at build time, so the loop remains reproducible without a neural model and the refusal threshold stays backend-aware.
- Harvesting of DIKSHA assets is gated pending bulk-download and offline-redistribution licensing.

rag-compliance-auditor github.com/aakash2410/MCP_RAG_Compliance

Sole-authored open-source package, published on PyPI.

- MCP server that certifies RAG and AI pipelines against the EU AI Act, GDPR/DPDP, HIPAA, SEBI/RBI and NIST AI RMF across 7 audit dimensions.
- Issues RS256-signed JWT certificates, exposes 8 MCP tools, and runs as a CI/CD gate with SARIF reporting.

PUBLICATIONS AND POLICY WRITING

Book chapter. Sangani, A., Doshi, P., & Sawant, V. "Unusual Social Media Behavior Detection Using Distributed Data Stream Mining." In *Practical Data Mining Techniques and Applications*. Auerbach Publications (CRC Press), 2022–23. Clickstream-based detection of hacked or bot-controlled accounts: users clustered by behaviour, a Very Fast Decision Tree per cluster, parallelised on Apache Spark, with concept-drift handling so genuine behavioural change retrains the model rather than raising a false alarm.

Preprint. Sangani, A. *Inverse-Objective Regulator.* arXiv system paper, sole author.

Policy. Section on AI incident-reporting mechanisms for India, in *AGI Governance*, written with the Office of the Principal Scientific Adviser to the Government of India, drawing on the EU AI Act and California's SB-53.

EVALUATION AND ASSESSMENT DESIGN

- Designed the evaluation framework for the **VYOMA Innovation Challenge** (₹80 lakh, run by Digital India Bhashini, Current AI and Kalpa Impact): jury rubric, use-case judging matrix, and the scoring thresholds applied through to the finale at IIIT-Delhi.
- Verified edge-AI performance claims submitted to the Gates Foundation, and wrote the reference architecture for AI-enabled coordinated care over ABDM-connected health systems.

SELECTED ACADEMIC PROJECTS

Baby Cry Reason Detection 2022 – 2023

Final project, team of 3. Guide: Prof. Prachi Tawde

- Audio classifier for the cause of an infant's cry (belly pain, burping, discomfort, hunger, tiredness) on a 457-sample dataset, using temporal, spectral, cepstral and prosodic features (MFCCs, spectral contrast, pitch).
- Addressed class imbalance with SMOTETomek; final model a soft-voting SVM + KNN ensemble, deployed in a Flutter app with a feedback loop for retraining.

Personality Prediction with CV and Social Media Analysis 2022

Team project, team of 4. Guide: Neha Ram

- Combined CV text analysis with Facebook behavioural data (myPersonality: 250 users, 9,917 status updates) to predict the Big Five traits. A 7-layer LSTM reached 0.82–0.85 accuracy, against 0.45–0.61 for SVM, logistic regression, gradient boosting and XGBoost baselines.

PROFESSIONAL EXPERIENCE

AI Consultant Feb 2026 – present

Kalpa Impact Mumbai

- Digital public infrastructure and AI governance consulting for UNDP, DPGA, the Gates Foundation and Co-Develop.
- Quantised the speech and vision models to INT8 via ONNX and restructured inference into parallel batched streams on Suno Sutra: peak memory down 40%, end-to-end latency from ~8s to 2–3s.

Business Technology Solutions Associate → Senior Associate Sep 2023 – Jan 2026

ZS Associates Pune

- Launched an agentic AI analytics product for natural-language querying of sales-performance metrics. Owned incentive-compensation product logic for a Fortune 100 pharma client at 100% accuracy. Led a team of 4.

Product Analyst Intern Jul 2022 – Feb 2023

Bright Money Bangalore

- Built A/B testing infrastructure and SQL dashboards. Launched a referral programme that saved $120K in customer-acquisition cost over 5 months.

TECHNICAL SKILLS

Methods: Bayesian inverse reinforcement learning, LLM evaluation and red-teaming, algorithmic fairness (AIF360, SHAP), Heckman-style selection models, A/B testing, RAG, MCP servers, INT8 quantisation and edge inference

Frameworks: EU AI Act, NIST AI RMF, ISO 42001, GDPR/DPDP, HIPAA, OWASP ASI, MITRE ATLAS

Tools: Python, SQL, ONNX, PyTorch-adjacent tooling (sentence-transformers), TensorFlow, scikit-learn, pandas, PostgreSQL, Apache Spark, Docker, Ansible, AWS

HONOURS

- Merit-based scholarship, 100% tuition waiver, D. J. Sanghvi College of Engineering (2019 – 2023).
- Recognised as a top 1% performer firm-wide at ZS Associates; promoted to Senior Associate within 8 to 10 months of joining.